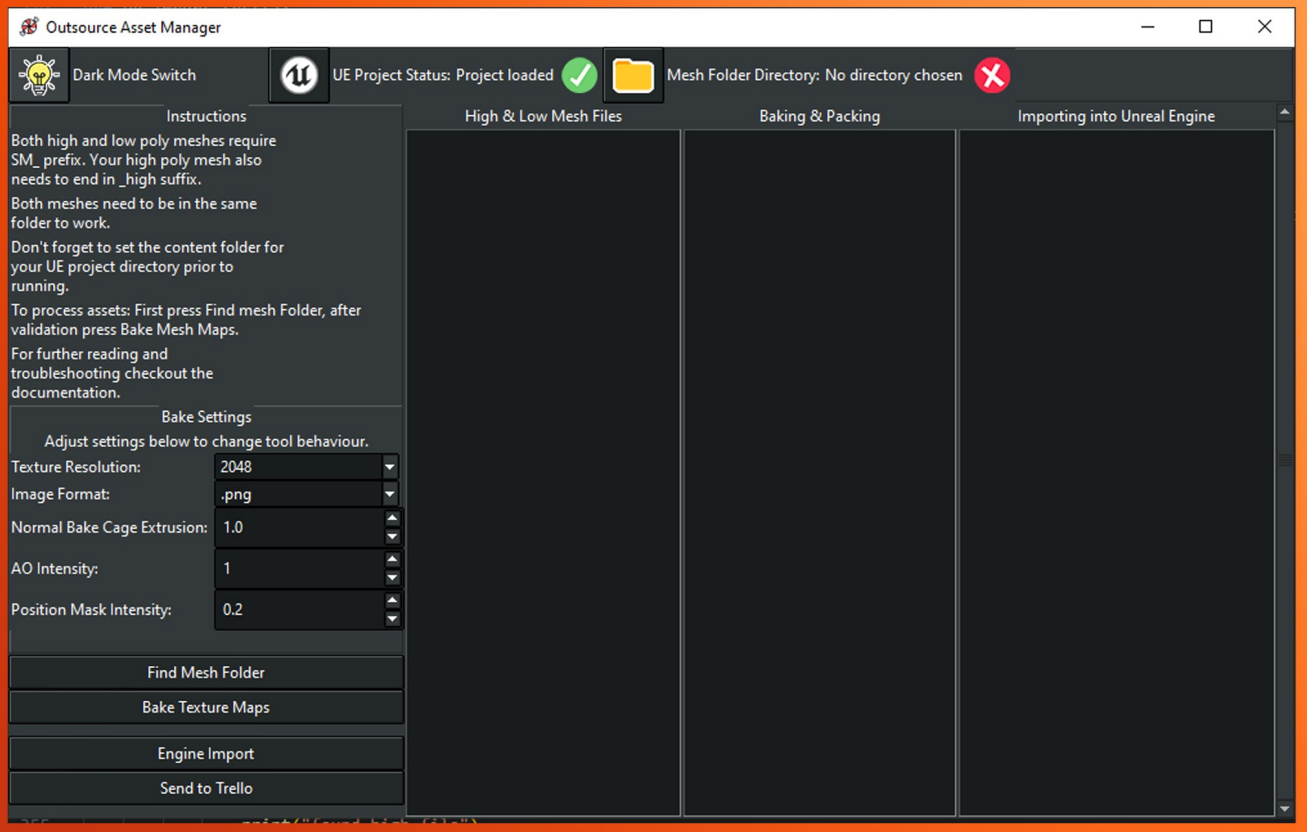
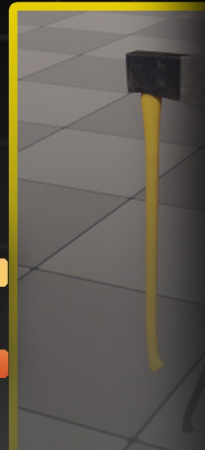
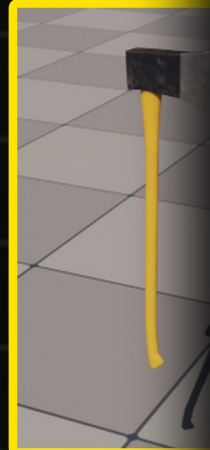
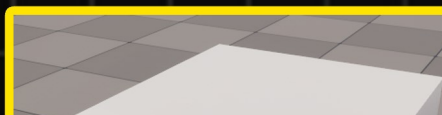




# Outsource-Asset Manager

By Jesse Olchawa

# Contents

|                         |          |
|-------------------------|----------|
| Tool Overview           | -----3   |
| Setting Up Configs      | - ---- 4 |
| Using the Tool          | - ---- 6 |
| Baking Textures         | -----7   |
| Auto Engine Import      | ----- 8  |
| Creating Thumbnails     | -----9   |
| Trello Progress Updater | -----10  |

# Tool Overview

## Tool Design:

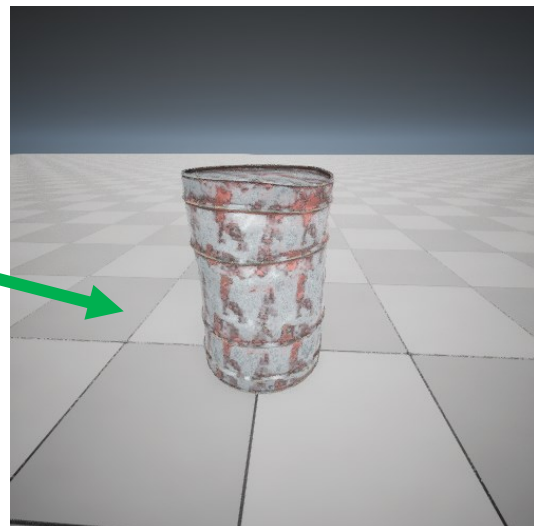
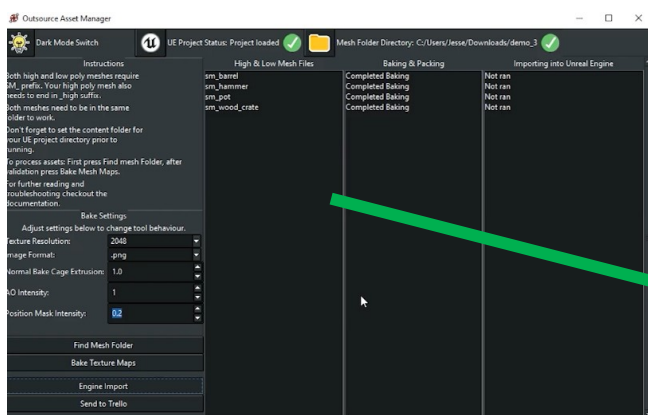
This tool is designed resolve baking, material generation, mesh engine setup for modular background assets. It can handle models with multiple materials and relies on shaders to create adjustable material instances in engine. **This tool is not intended to be used on bespoke, highly detailed hero assets, its better to get an artist to author unique textures for those models to add narrative detail.**

## This tool works in 3 key stages:

- Importing and validating high and low poly meshes
- Baking and mask map generation using Blender
- Importing and final setting up in Unreal Engine

**There is also a optional stage for generating thumbnails for chosen assets in Unreal Engine using the widget if you also have setup the Trello config file.** Thumbnails will be uploaded to a new card with the mesh name and timestamp date for easier task management for review.

**This tool requires proper setup via its config files, do not miss this step as it affects baking results and engine importing. Let's set them up.**



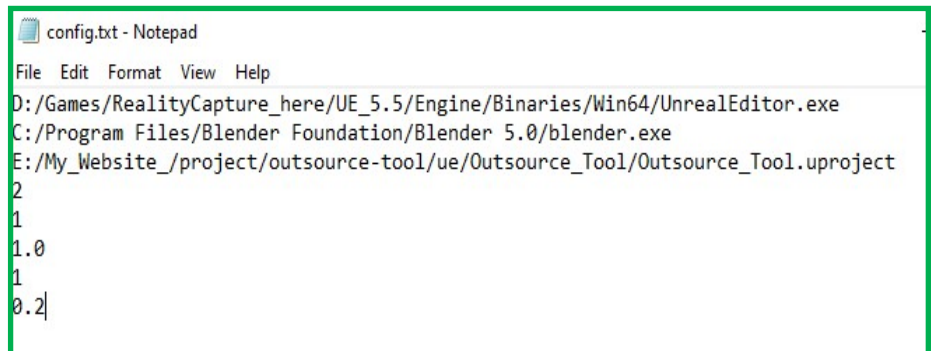
# Setting Up Configs

## Primary Config — config.txt:

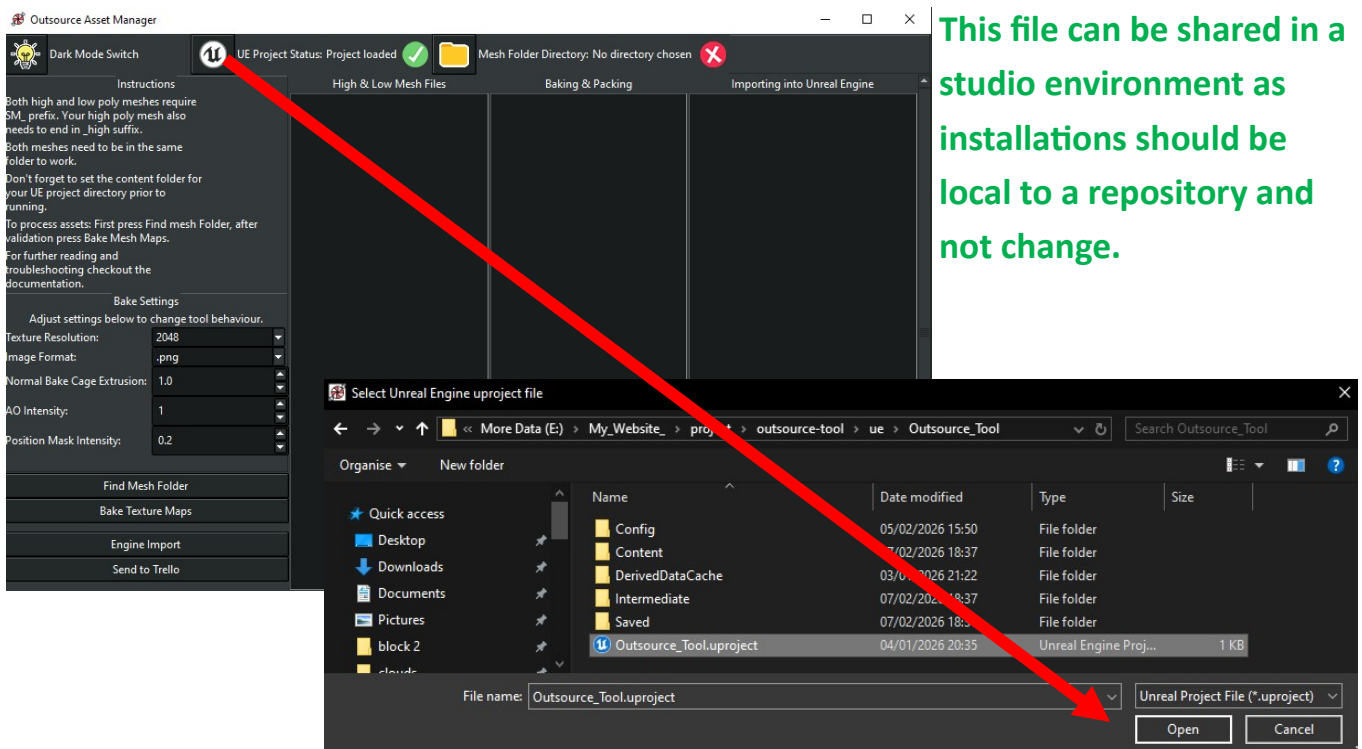
This config can be found in /python\_script\_loc/shader\_graphs. If it is missing or invalid, on opening the outsource tool it will be regenerated. Your config needs to contain file locations for installations of Blender, Unreal Engine and project file location.

### Here is a example config.txt:

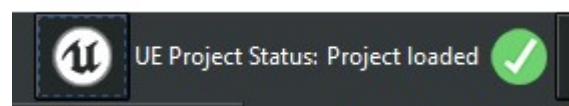
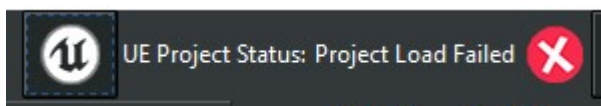
- 1) Line 1 needs to have the UnrealEditor.exe
- 2) Line 2 needs to have Blender 5.0 or higher blender.exe
- 3) Line 3 needs to have the Unreal Project .uproject location. **This can also be set in the tool and will be written to the file using the project buttons:**



```
config.txt - Notepad
File Edit Format View Help
D:/Games/RealityCapture_here/UE_5.5/Engine/Binaries/Win64/UnrealEditor.exe
C:/Program Files/Blender Foundation/Blender 5.0/blender.exe
E:/My_Website_/project/outsource-tool/ue/Outsource_Tool/Outsource_Tool.uproject
2
1
1.0
1
0.2
```



If locations are set correctly this config will showcase a green check mark and status.



### Material Locations and Unreal Folder Structure — mat\_data.csv:

This csv file will be generated as a empty template if not found however requires manual filling in. **It can be duplicated in a studio environment as it remains consistent.**

**Here's what to add:**

- 1) Line 1 is the name of the material based on the material ids assigned in 3D software.
- 2) Line 2 is the location of the master material that uses that name in engine, based on the /Game/ folder as root. Ensure the location matches the row above.
- 3) Line 3 is the location to import textures into the project.
- 4) Line 4 is the location to import meshes into the project.
- 5) Line 5 is the location to create material instances into the project.

**Here is a example of a correctly filled out mat\_data.csv file.**

|   | A                                 | B                                   |
|---|-----------------------------------|-------------------------------------|
| 1 | m_wood                            | m_metal                             |
| 2 | /Game/Demo/Material/m_wood.m_wood | /Game/Demo/Material/m_metal.m_metal |
| 3 | /Game/Demo/Textures               |                                     |
| 4 | /Game/Demo/Meshes                 |                                     |
| 5 | /Game/Demo/Material               |                                     |
| 6 |                                   |                                     |

### Optional Trello Integration — trello\_config.txt:

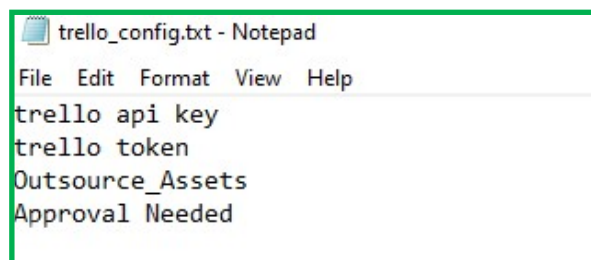
If you want to be able to use the Trello thumbnail widget later on, you need to fill out the config to connect your Trello account and boards together to the tool.

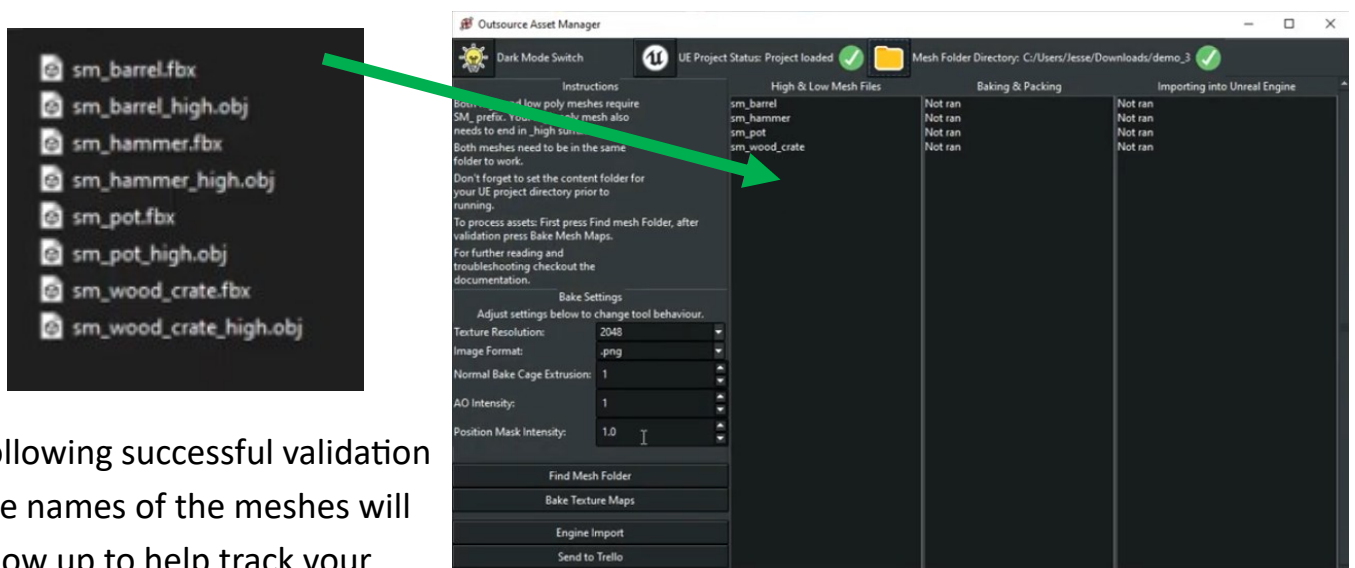
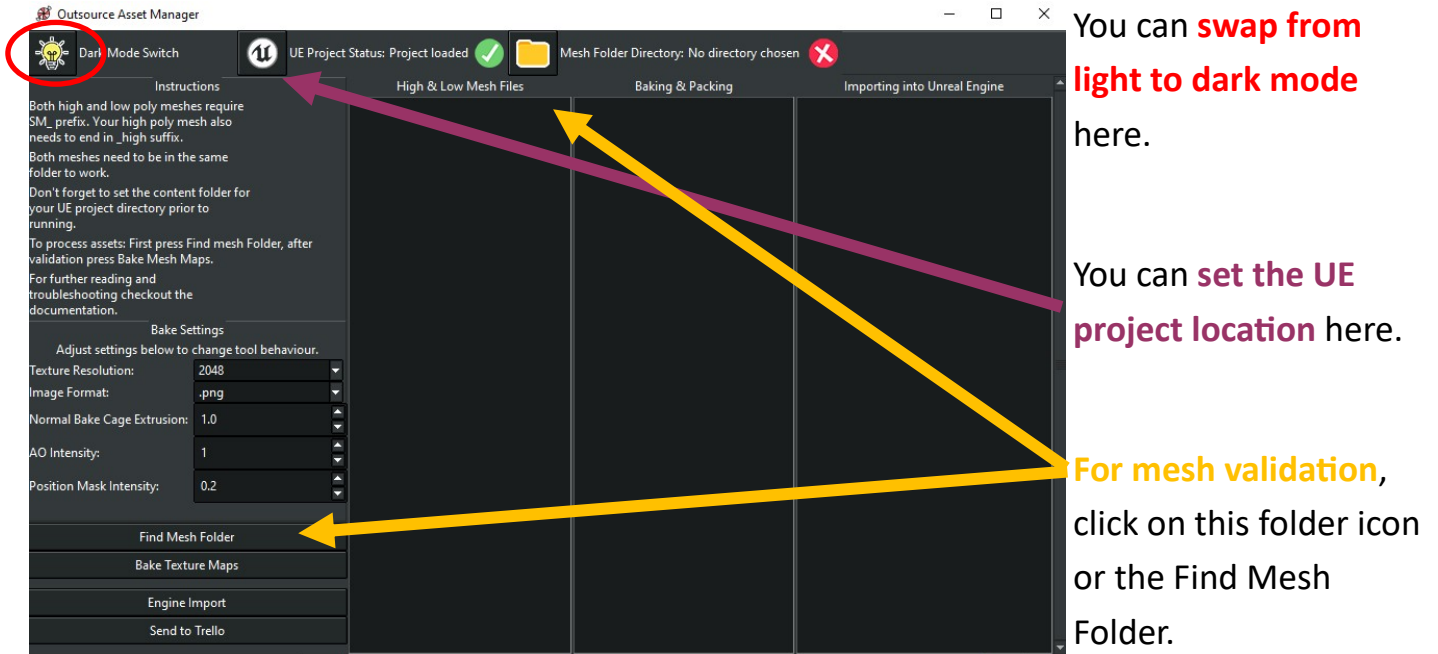
**Here's what to add:**

- 1) Line 1 is the Trello API key for the board.
- 2) Line 2 is the Trello Token key for the account.
- 3) Line 3 is the board name, e.g. Outsource Assets
- 4) Line 4 is the category to post into the board for example Approval Needed

**Here is a example of a correctly filled out trello\_config.txt minus the tokens.**

**Never reveal your token! For better security, ensure the studio has a separate account used for progress management posting only.**

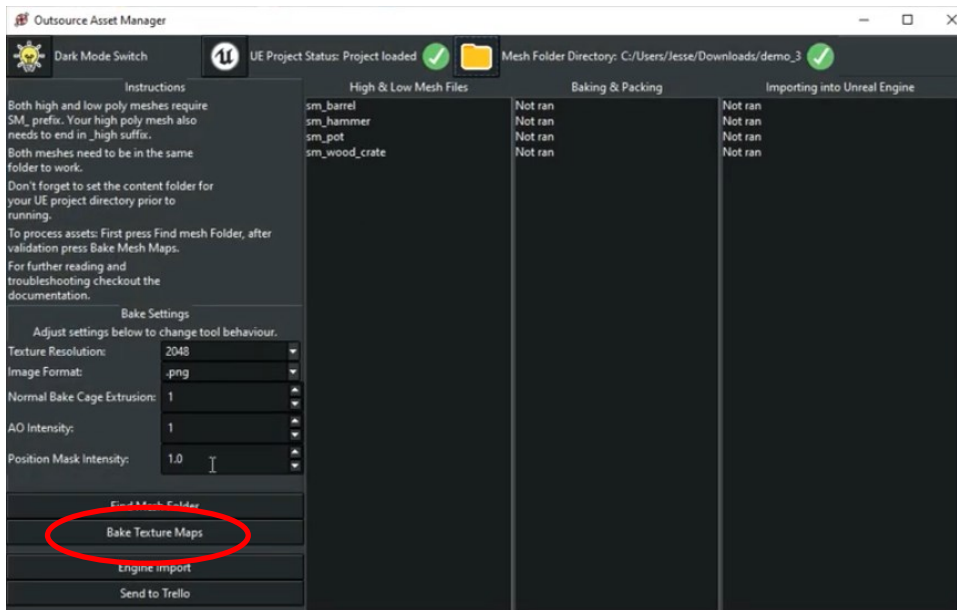




Following successful validation the names of the meshes will show up to help track your progress. **Let's move onto baking next.**

# Baking Textures

## Step 2 — Baking Textures:



Now the tool has found both high and low poly meshes it can bake out normal map and the channel packed mask map (\_AOCEP).

Adjust settings prior to baking such as the format of the file (.TGA) or texture resolution.

Click Bake Texture Maps when ready, your progress will update and you will see files begin to appear inside a folder called after the mesh. **Here is a example of a successful bake.**

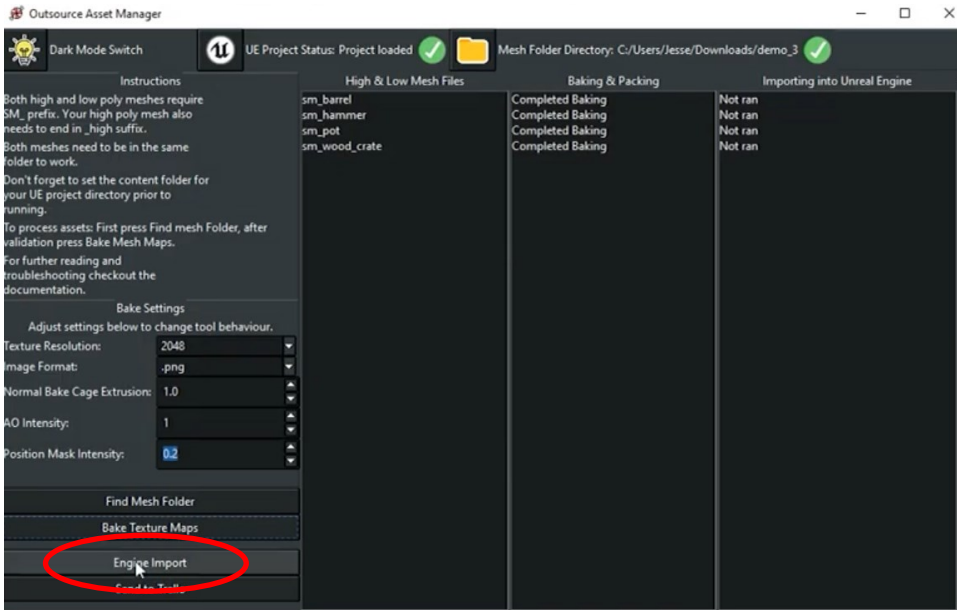


After adjusting settings you can click Bake Texture Maps again to recreate these textures and refresh them in real-time. **Warning this will rebake all of the maps currently loaded in via mesh names, so if you only want to rebake select meshes consider re-validating a folder with only the single mesh loaded in.** Let's move onto engine importing next.



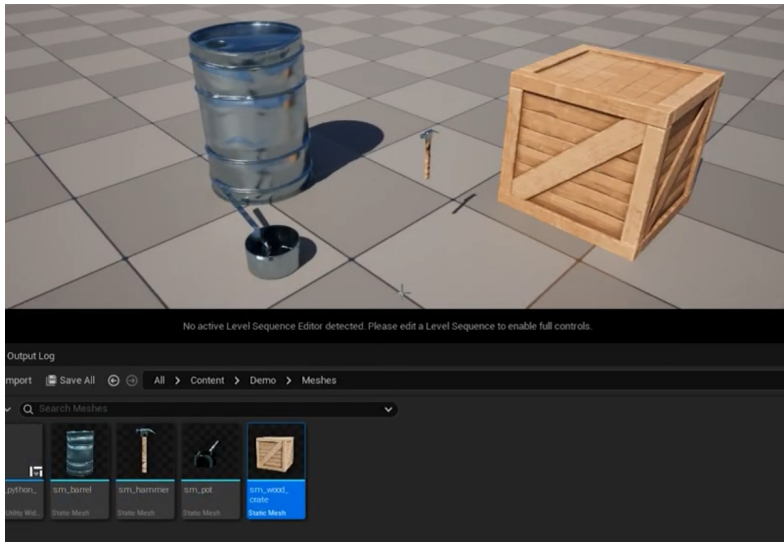
# Auto Engine Import

## Step 3 — Engine Import:



Now we have mesh maps baked we can auto import and setup our meshes with material instances in our project scene. **Make sure your UE repository is linked and green check is displayed above here to correctly connect and import meshes in.**

**Ensure Unreal Engine is closed, do not have it open during importing. When ready click on the Engine Import button.** The terminal will flash briefly however it can be ignored.



Now you can open up your Unreal Engine Project and see the meshes have been setup with material instances matching mesh name.

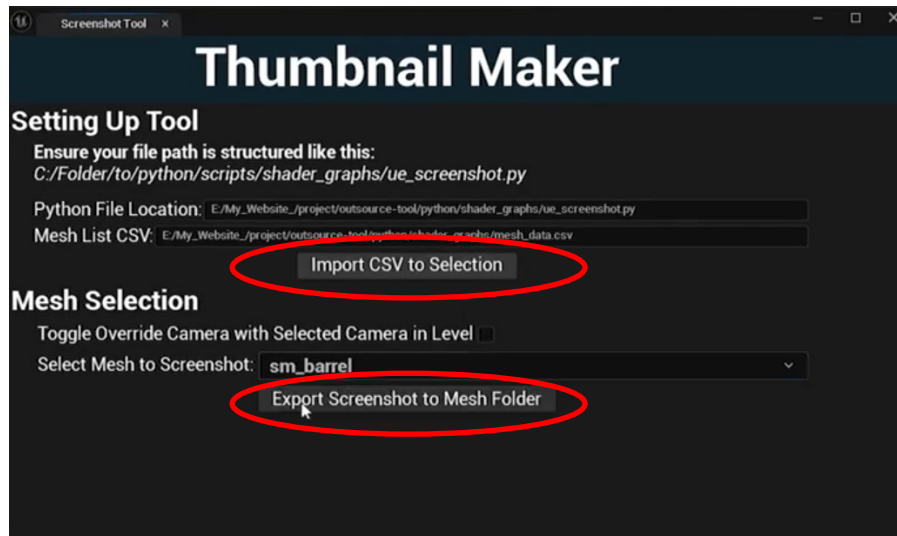
**Adjust the material instances on the meshes to fit your concept art/reference.** Here are some examples:



**The amount of editability is linked to your master material, consider exposing more parameters for colour editing or mask manipulation to get different outlooks of wear and tear.** Keep graphs simple and cheap, these are background assets after all.

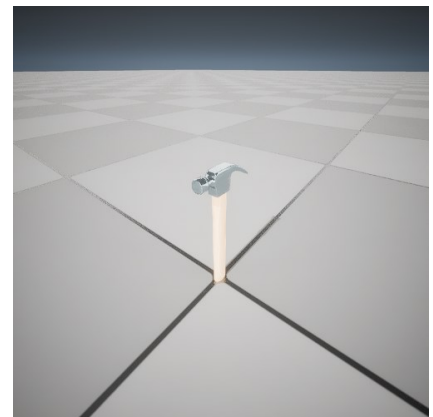
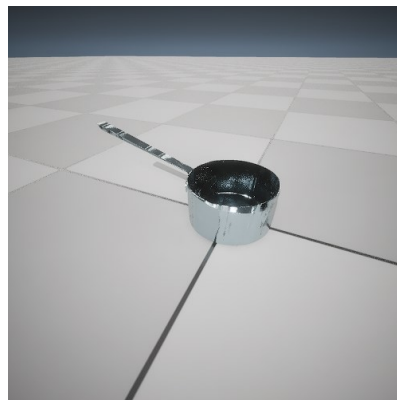
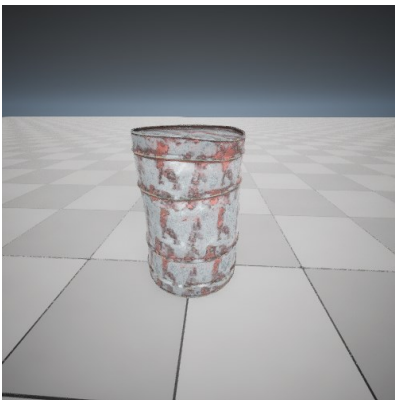
# Creating Thumbnails

## Using the Thumbnail Widget)



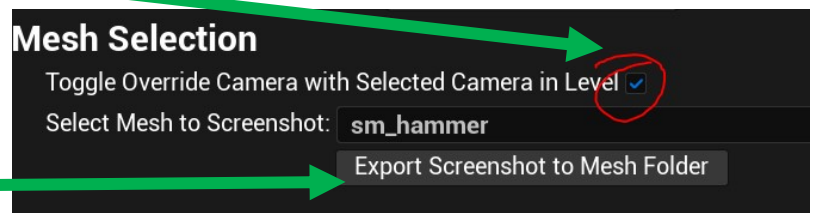
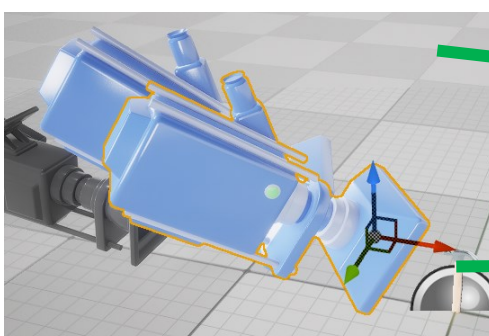
Right click on the Widget in Content Browser to load it. Set the python file path and the mesh list.csv file which is created under shader\_graphs folder of the python script. This contains validated mesh information. **After importing the mesh selection dropdown will become available.**

Select the mesh you want to create a screenshot of and **click Export Screenshot to Mesh Folder to take an automatic screenshot** in the screenshot level (it will auto load). Check the original mesh import folder to see the screenshots. **Some examples of thumbnails:**

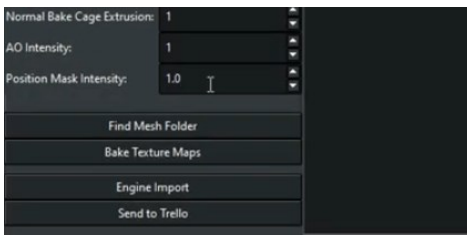


If you want to take a custom screenshot from a different angle/mesh, toggle on the override button and select the camera in the level before clicking the export button again.

**This will now take a screenshot using that cameras viewport and export it to the folder.**



# Trello Progress Updater



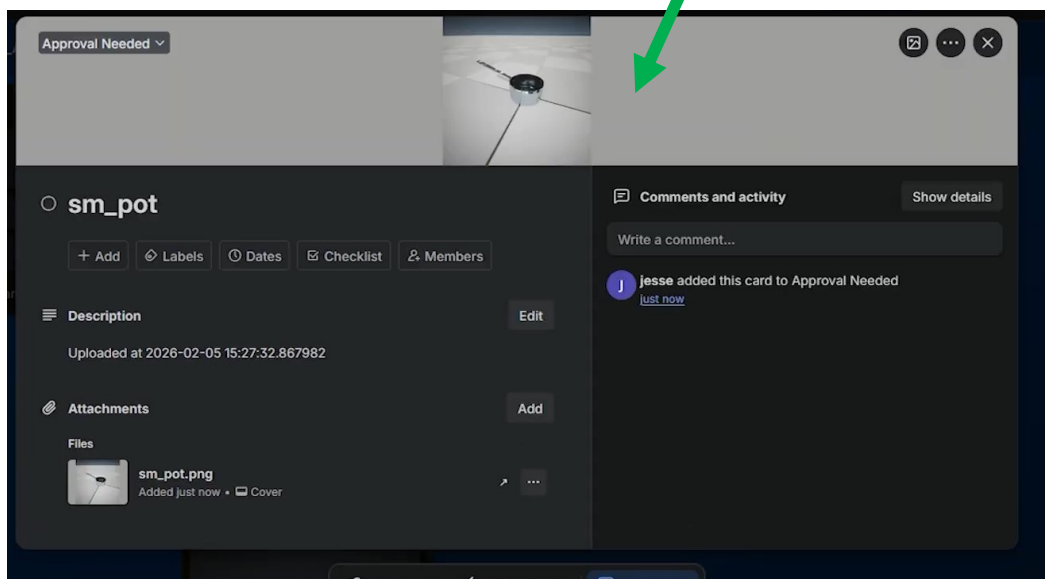
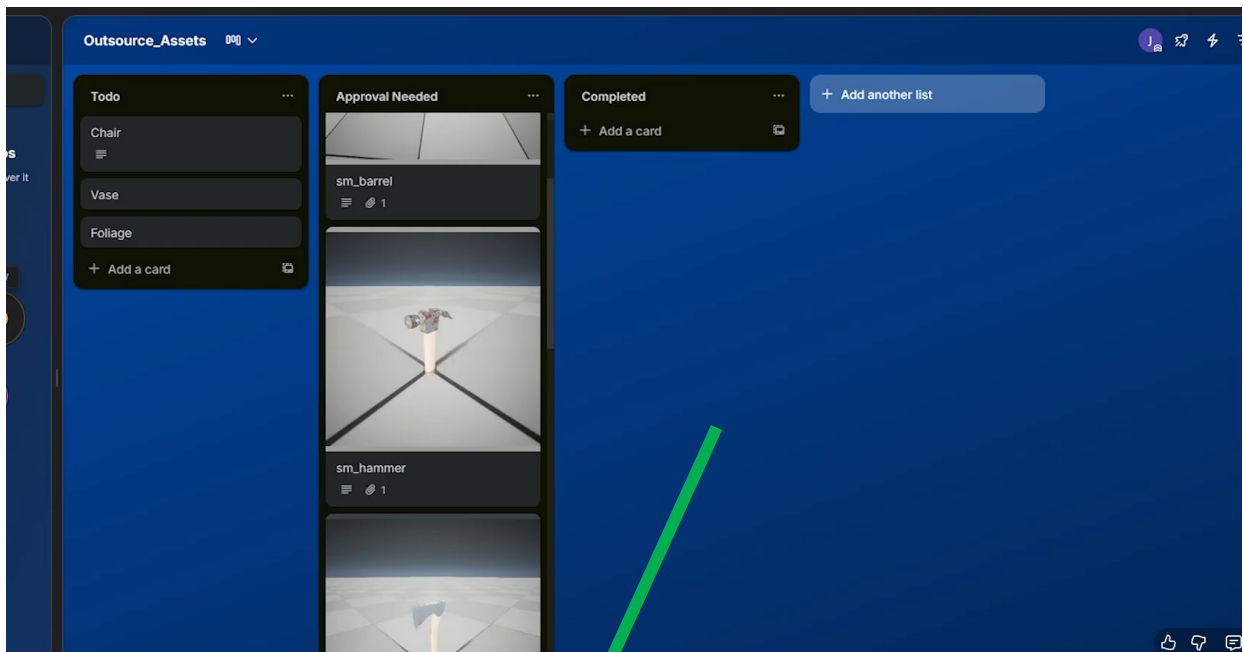
## Sending to Trello)

After thumbnails are made you can click the Send to Trello button as long as the config Trello file has been setup successfully. This will attempt to run through all the listed meshes in the process list.



Following a successful process to Trello a status message will appear and you can check the specified board e.g. Approval Needed for the card.

**Example below:**



The card contains the mesh name, thumbnail and the date uploaded into description. **You can further edit these cards and sort them into the correct lists as required by project leads.**